NAME: _____ Geometry (FL B.E.S.T.)

3.6 Transformations and Congruence

The modern definition of congruence is built entirely on rigid motions: two figures are congruent exactly when some sequence of translations, reflections, and rotations maps one perfectly onto the other. This is more powerful than just comparing measurements, because it explains *why* corresponding parts must match (CPCTC) and it works even when the mapping needs two or three steps chained together. This section proves congruence with explicit coordinates, contrasts congruence with dilation (which changes size and breaks the rigid-motion definition), and reverses the process to find the transformation sequence from two given triangles.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. Congruent means same size and what?

2. List three rigid motions.

3. Rigid motion maps $\triangle ABC$ to $\triangle XYZ$. Conclude?

4. $\triangle ABC \cong \triangle DEF$. $\overline{BC} \cong$?

5. $\triangle ABC \cong \triangle DEF$. $\angle A \cong$?

6. Is dilation (factor 2) a congruence?

7. Congruence symbol?

8. Corresponding parts of congruent triangles are what?

9. Can a reflection alone establish congruence?

10. Can a translation alone establish congruence?

11. Must congruent figures have equal corresponding side lengths?

12. Must congruent figures have equal corresponding angles?

13. Translate then rotate a triangle. Congruent to original?

14. $\triangle ABC$ has $A(0,0)$, $B(5,0)$, $C(0,2)$. $\triangle DEF$ has $D(0,0)$, $E(-5,0)$, $F(0,-2)$. Find the single rigid motion mapping $\triangle ABC$ to $\triangle DEF$.

15. $\triangle PQR$ has $P(1,2)$, $Q(4,2)$, $R(1,5)$. Its image is $P'(-1,-1)$, $Q'(-4,-1)$, $R'(-1,2)$. Find a sequence of two rigid motions that maps $\triangle PQR$ to $\triangle P'Q'R'$.

✓ Answer Key

1. Shape
2. Translation, reflection, rotation
3. $\triangle ABC \cong \triangle XYZ$
4. $\overline{EF}$
5. $\angle D$
6. No
7. $\cong$
8. Congruent
9. Yes
10. Yes
11. Yes
12. Yes
13. Yes
14. $180°$ rotation about the origin (rule $(x,y) \to (-x,-y)$)
15. Reflect over the y-axis, then translate by $\langle 0,-3\rangle$ (check: $P(1,2) \to (-1,2) \to (-1,-1)$ ✓; $R(1,5) \to (-1,5) \to (-1,2)$ ✓)